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**CS 441 - HW 2: Trees and MLPs**

Complete the claimed points and sections below.

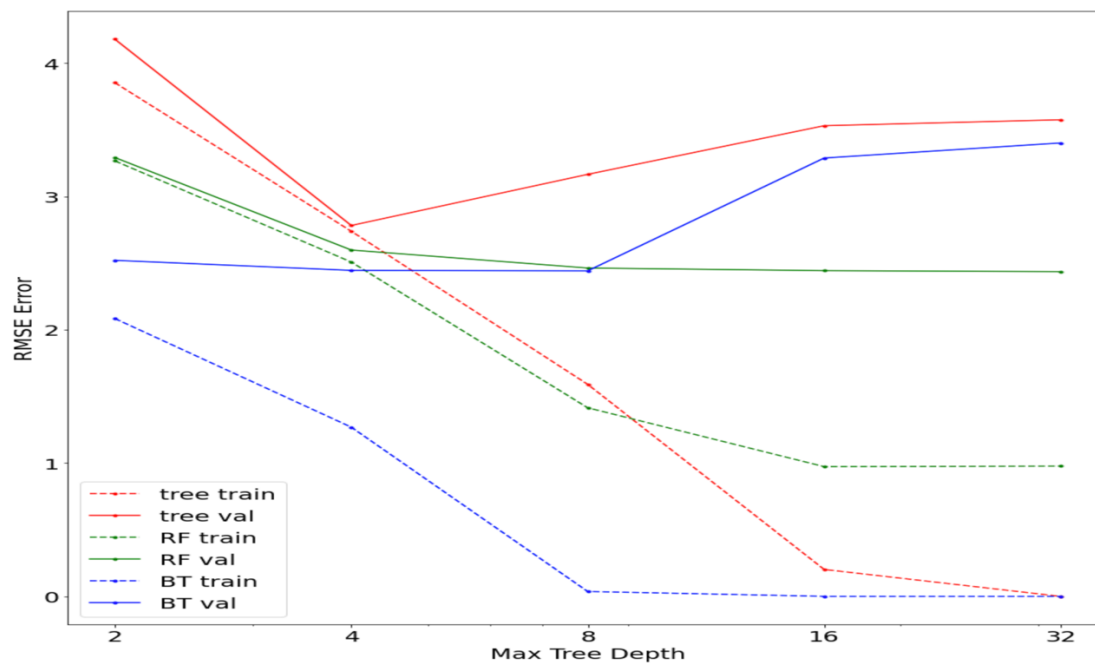
**Total Points Claimed**

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- |                                          |           |
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| 3. Species Prediction                    |           |
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## 1. Model Complexity with Tree Regressors

Include your plot below.



Analyze your results:

1. For a given max tree depth, which regressor model (tree, RF, BT) has the lowest bias (or is most powerful)?

**For a given max tree depth, the Random Forest Regressor model has the lowest bias (in other words, the Random Forest Regressor model is the most powerful).**

2. For regression trees, what tree depth achieves minimum validation error?

**For regression trees, tree depth 4 achieves the minimum validation error (tree depth 8 is a close second, but tree depth 4 definitely achieves the minimum validation error).**

3. A model “overfits” when increasing the complexity increases the validation error. Which model is least prone to overfitting? Why?

**The Random Forest Regressor model is least prone to overfitting. I say this because, for the Random Forest Regressor model, as we increase the complexity (Max Tree Depth), the validation error is decreasing (or in some cases, it stays the same, but it never increases). For the other two models, the validation error increases, or it sometimes stays the same, but it rarely decreases (so, the other two models are more prone to overfitting).**

4. Do boosted trees seem to perform better with smaller or larger trees? Why?

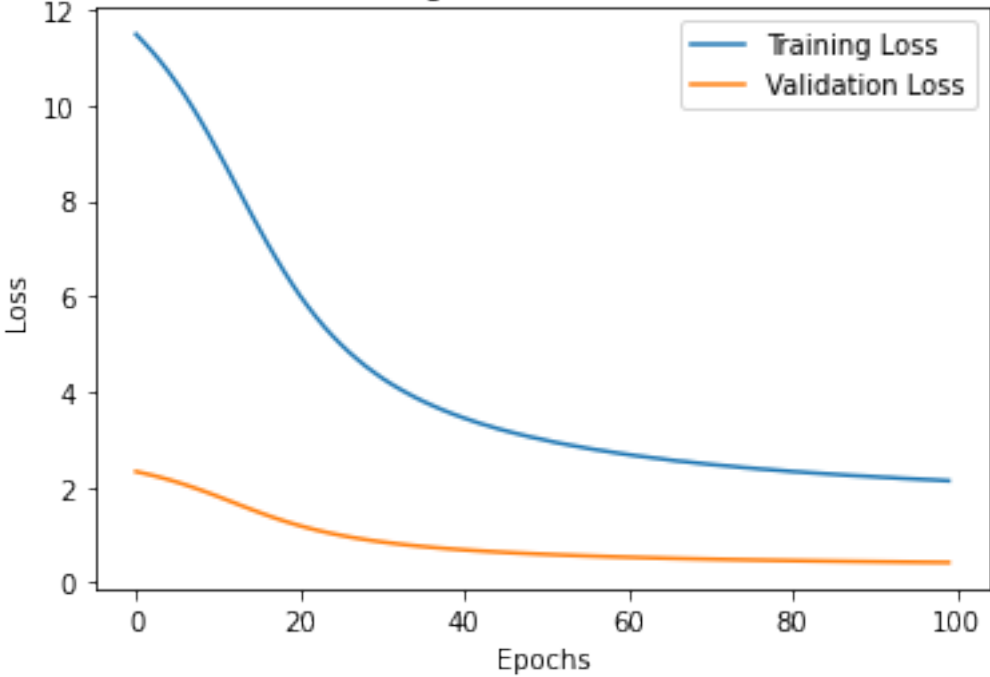
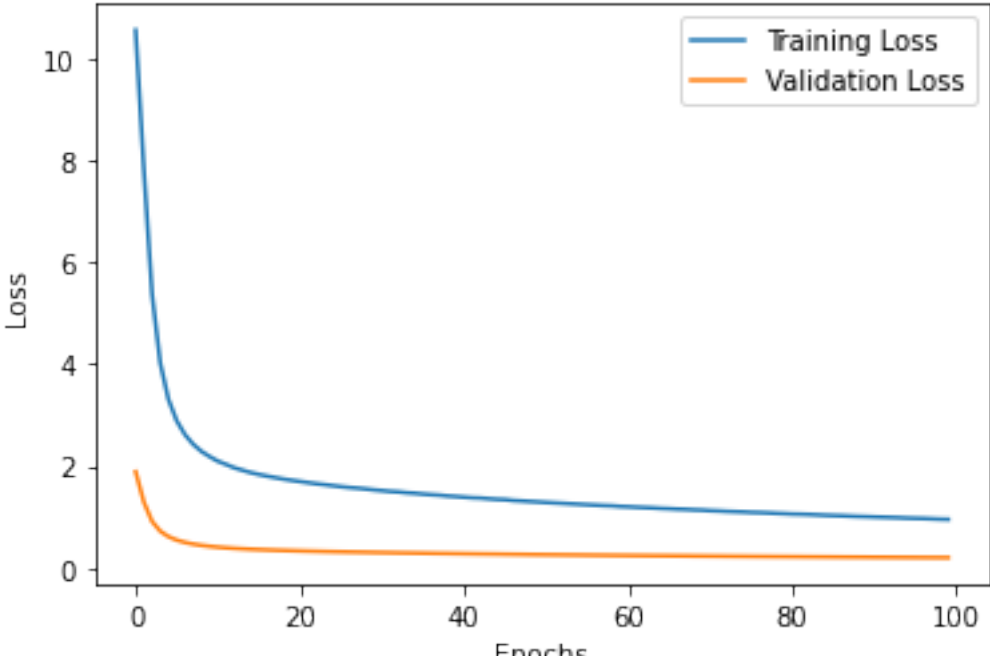
**Boosted trees seem to perform better with smaller trees. I say this because, when it comes to boosted trees, using too many trees or too large trees could lead to overfitting. Also, as we can see the boosted trees in the plot above, when the max tree depth is 16 or 32, we can see a pattern of overfitting.**

## 2. MLPs with MNIST

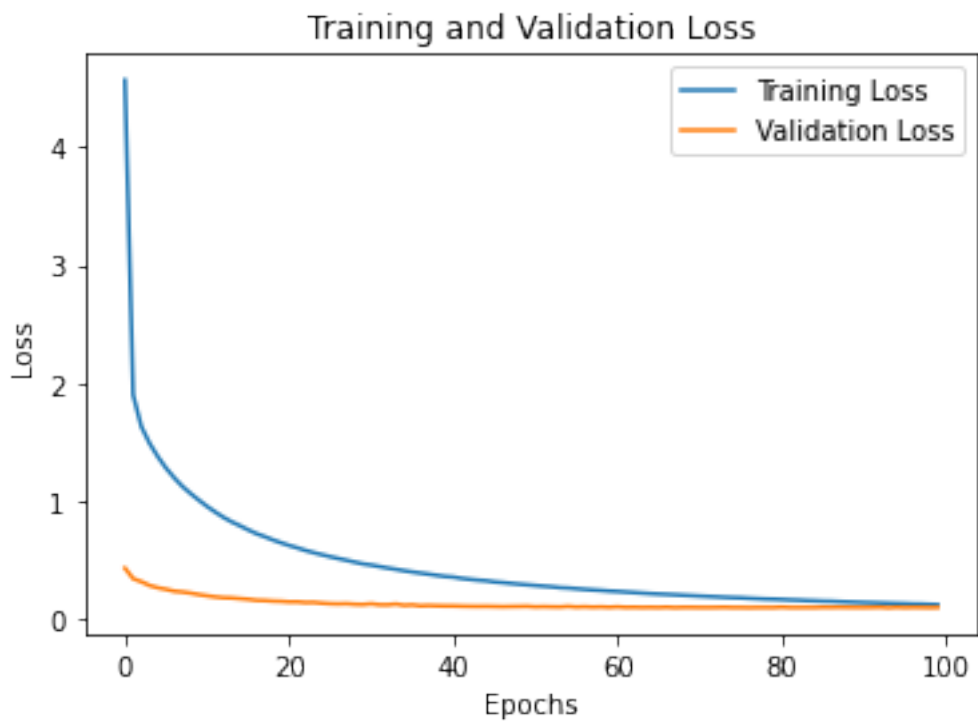
Using the default parameters, report your training and validation losses after the given number of epochs

| # training epochs | Training Loss       | Validation Loss     |
|-------------------|---------------------|---------------------|
| 25                | 0.5433568798769743  | 0.13234645261978492 |
| 50                | 0.2882781217877681  | 0.10586874272960883 |
| 100               | 0.11961032705715834 | 0.09638118820312695 |

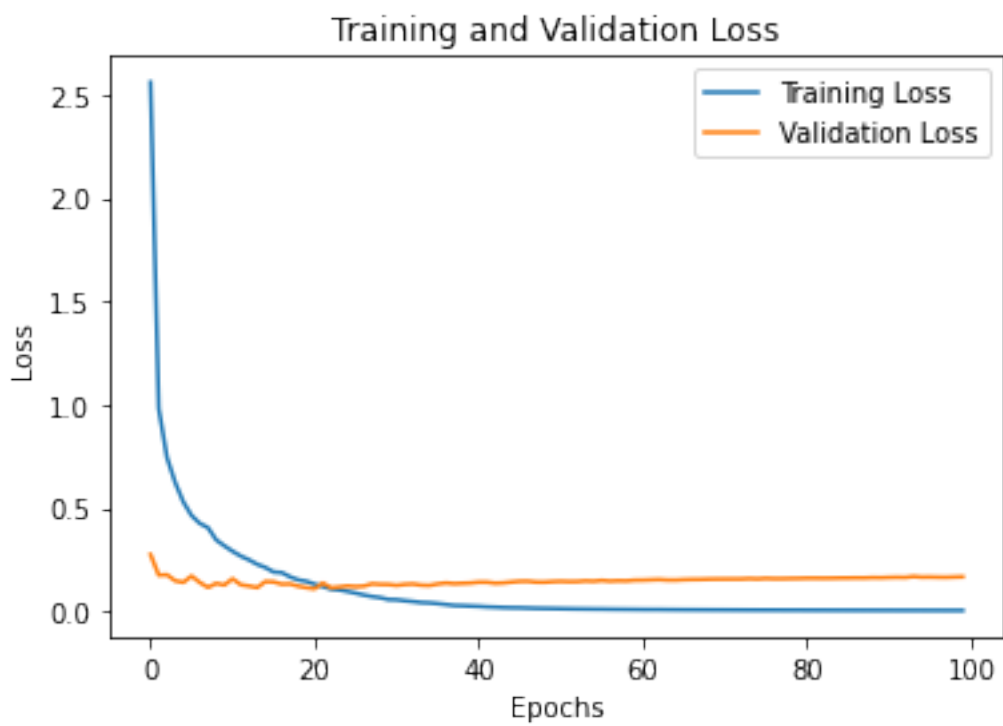
Show the loss curves for at least 3 different learning rates

| Learning Rate         | Loss Curve Plot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                 |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------|-----------------|---|------|-----|----|-----|-----|----|-----|-----|----|-----|-----|----|-----|-----|-----|-----|-----|
| Learning Rate = 0.001 | <p>Training and Validation Loss</p>  <p>Loss</p> <p>Epochs</p> <p>Training Loss<br/>Validation Loss</p> <table border="1"><caption>Approximate data points for Learning Rate = 0.001</caption><thead><tr><th>Epochs</th><th>Training Loss</th><th>Validation Loss</th></tr></thead><tbody><tr><td>0</td><td>11.5</td><td>2.5</td></tr><tr><td>20</td><td>6.5</td><td>1.5</td></tr><tr><td>40</td><td>3.8</td><td>0.8</td></tr><tr><td>60</td><td>2.8</td><td>0.6</td></tr><tr><td>80</td><td>2.4</td><td>0.5</td></tr><tr><td>100</td><td>2.2</td><td>0.5</td></tr></tbody></table> | Epochs          | Training Loss | Validation Loss | 0 | 11.5 | 2.5 | 20 | 6.5 | 1.5 | 40 | 3.8 | 0.8 | 60 | 2.8 | 0.6 | 80 | 2.4 | 0.5 | 100 | 2.2 | 0.5 |
| Epochs                | Training Loss                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Validation Loss |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| 0                     | 11.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2.5             |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| 20                    | 6.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1.5             |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| 40                    | 3.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.8             |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| 60                    | 2.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.6             |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| 80                    | 2.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.5             |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| 100                   | 2.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.5             |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| Learning Rate = 0.01  | <p>Training and Validation Loss</p>  <p>Loss</p> <p>Epochs</p> <p>Training Loss<br/>Validation Loss</p> <table border="1"><caption>Approximate data points for Learning Rate = 0.01</caption><thead><tr><th>Epochs</th><th>Training Loss</th><th>Validation Loss</th></tr></thead><tbody><tr><td>0</td><td>10.5</td><td>2.0</td></tr><tr><td>20</td><td>1.8</td><td>0.5</td></tr><tr><td>40</td><td>1.4</td><td>0.4</td></tr><tr><td>60</td><td>1.2</td><td>0.3</td></tr><tr><td>80</td><td>1.1</td><td>0.3</td></tr><tr><td>100</td><td>1.0</td><td>0.3</td></tr></tbody></table> | Epochs          | Training Loss | Validation Loss | 0 | 10.5 | 2.0 | 20 | 1.8 | 0.5 | 40 | 1.4 | 0.4 | 60 | 1.2 | 0.3 | 80 | 1.1 | 0.3 | 100 | 1.0 | 0.3 |
| Epochs                | Training Loss                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Validation Loss |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| 0                     | 10.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2.0             |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| 20                    | 1.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.5             |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| 40                    | 1.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.4             |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| 60                    | 1.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.3             |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| 80                    | 1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.3             |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |
| 100                   | 1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.3             |               |                 |   |      |     |    |     |     |    |     |     |    |     |     |    |     |     |     |     |     |

Learning Rate  
= 0.1



Learning Rate  
= 1



Report the losses and errors for your best model here:

**I achieved the improved model by changing the learning rate. On the original learning rate (0.1), after 100 epochs, the training loss is 0.11961032705715834, and the validation loss is 0.09638118820312695. Here is the new training loss and the new validation loss after changing the learning rate to 0.12 (after 100 epochs):**

| Training Loss       | Validation Loss     |  |
|---------------------|---------------------|--|
| 0.09166320979308623 | 0.08955111010716511 |  |

**I could not figure out how to get the training, validation, and test errors.**

| Training Error | Validation Error | Test Error |
|----------------|------------------|------------|
|                |                  |            |

### 3. Species Prediction

#### a. Visualization of Features

Report your visualizations (scatterplots) of different features and the relationships.

| Features                                                                                                                                                                                                                                                                                                                                                                                     | Visualization                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Features:</b></p> <p>1: Culmen Depth</p> <p>2: Culmen Length</p> <p><b>Relationships:</b></p> <p>1: Here we can see that the culmen length usually increases as the culmen depth increases (for all three species).</p> <p>2: We can also see that the Chinstrap species have the highest culmen depth average, and the the Gentoo species have the highest culmen length average.</p> | <p>culmen depth vs culmen length (mm)</p> <p>culmen_length_mm</p> <p>culmen_depth_mm</p> <p>● Adelie<br/>● Chinstrap<br/>● Gentoo</p> |

|                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Features:</p> <p>1: Body Mass</p> <p>2: Culmen Length</p> <p>Relationships:</p> <p>1: Here we can see that all three species are very different based on the body mass/culmen length data.</p> <p>2: We can also see that, on average, the Adelie species have the smallest body mass and the smallest culmen length, when compared to the other two species.</p> | <p>body mass vs culmen length (mm)</p> <p>This scatter plot shows the relationship between body mass (x-axis, 180-220) and culmen length (y-axis, 40-60 mm) for three species. Adelie (cyan) is clustered at lower values (mass ~180-200, length ~35-45). Chinstrap (brown) is in the middle (mass ~185-210, length ~45-55). Gentoo (magenta) is at higher values (mass ~210-230, length ~45-55). There is some overlap between Chinstrap and Gentoo.</p> <p>Legend: Adelie (cyan), Chinstrap (brown), Gentoo (magenta)</p> |
| <p>Features:</p> <p>1: Flipper Length</p> <p>2: Culmen Depth</p> <p>Relationships:</p> <p>1: Here we can see that the culmen depth usually increases as the flipper length increases (for all three species).</p> <p>2: We can also see that the flipper length/culmen depth data for the Adelie species and the Chinstrap species is very similar.</p>              | <p>flipper length vs culmen depth (mm)</p> <p>This scatter plot shows the relationship between flipper length (x-axis, 180-220) and culmen depth (y-axis, 15.0-20.0 mm) for three species. Adelie (red) and Chinstrap (green) are clustered together at higher values (flipper ~180-200, depth ~16-20). Gentoo (cyan) is at lower values (flipper ~210-230, depth ~14-16). There is significant overlap between Adelie and Chinstrap.</p> <p>Legend: Adelie (red), Chinstrap (green), Gentoo (cyan)</p>                     |

You may extend the table if you have more results

If you had to choose two features to make your species classification based on this analysis, which features would you choose?

**The two features that I would choose to make my species classification are body mass and flipper length. I say this because I think that all of the other features depend on these two features a lot. For example, if the body mass or flipper length value changes, most of the values for the other features will also change.**

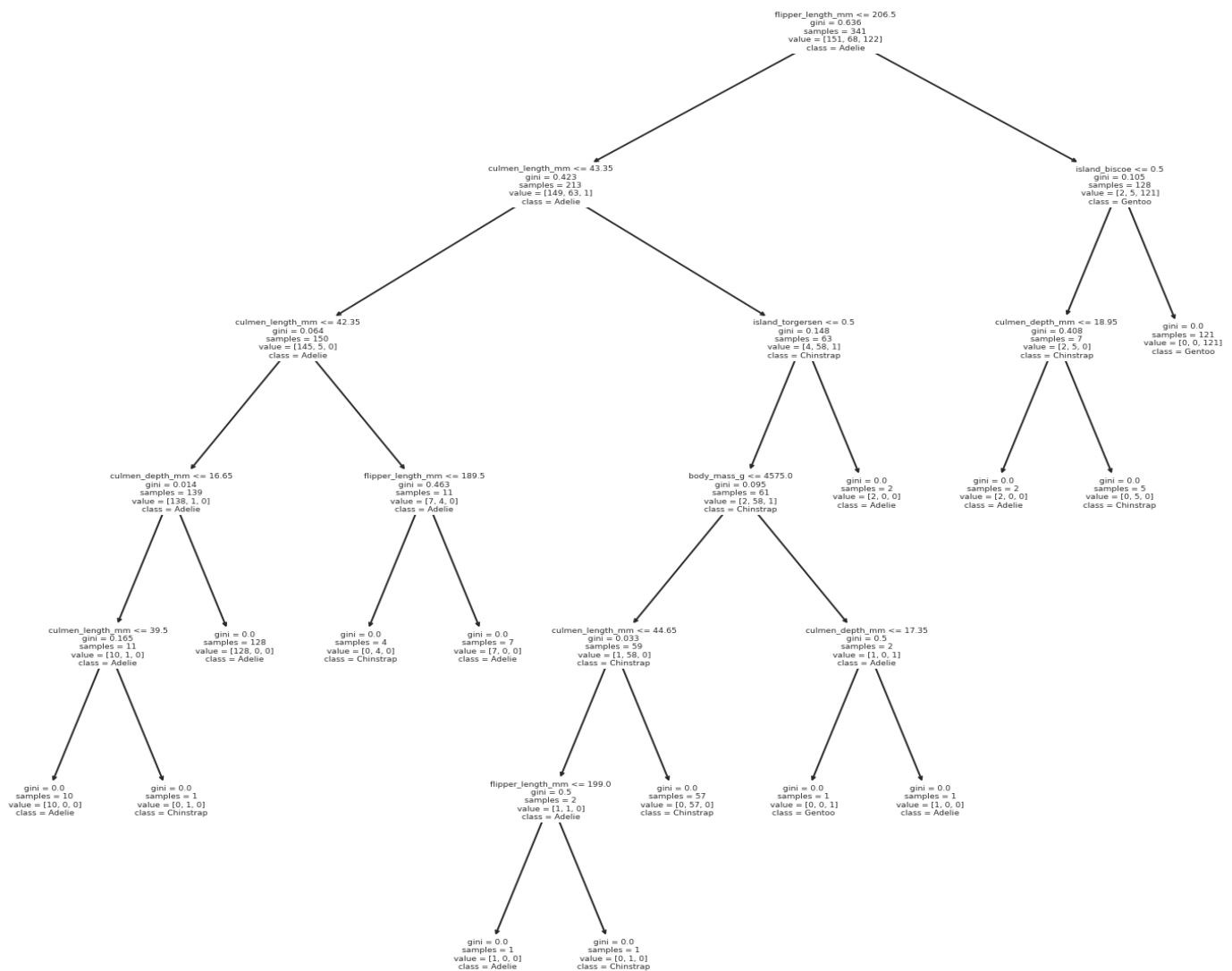
## b. Simple rule to identify Gentoo

Write down the simple two-part rule to identify Gentoo. Display your decision tree with labeled features and classes.

Here is my simple two-part rule to identify gentoo:

If flipper length is greater than 206.5 mm and island biscoe is greater than 0.5, then the species is gentoo. Otherwise it is not.

Here is my decision tree with labeled features and classes:



### c. Models

Describe the model that achieves best 5-fold cross-validation accuracy:

**The model that achieves the best 5-fold cross-validation accuracy is the Logistic Regression model. To achieve this, I started off by importing the Logistic Regression model from sklearn. Then I passed a few parameters to the Logistic Regression model. These parameters include the maximum number of iterations and the C value. The default C value, which is 1.0, also got a cross-validation accuracy above 99%, but making the C value equal to 4.0 improved it a little bit more. Lastly, I set the Logistic Regression model equal to the model variable and passed the model variable to the cross-validation score function.**

5-fold Cross-Validation Accuracy: 0.9970588235294118

### 3. Stretch Goals

#### a. Improve MNIST Classification Performance using MLPs

Report the classification val and test errors and details of your best method. Describe your approach and parameters. Feel free to change the MLP batch size, learning rate, number of epochs, hidden layer size, activation layer, or anything else.

**I am only claiming 10 points for this question because I could not figure out how to get the training, validation, and test errors. However, I was able to find another way, instead of changing the learning rate, to improve MNIST when it comes to training loss and validation loss.**

**Here is the original training loss and validation loss:**

**Training Loss: 0.11961032705715834**

**Validation loss: 0.09638118820312695**

**Here is the new training loss and validation loss after keeping everything the same except for the hidden layer size (the original hidden layer size is 64, and I doubled it to 128):**

**Training Loss: 0.09492775738143767**

**Validation loss: 0.08410414064732882**

#### b. Find a second simple rule to identify Gentoo

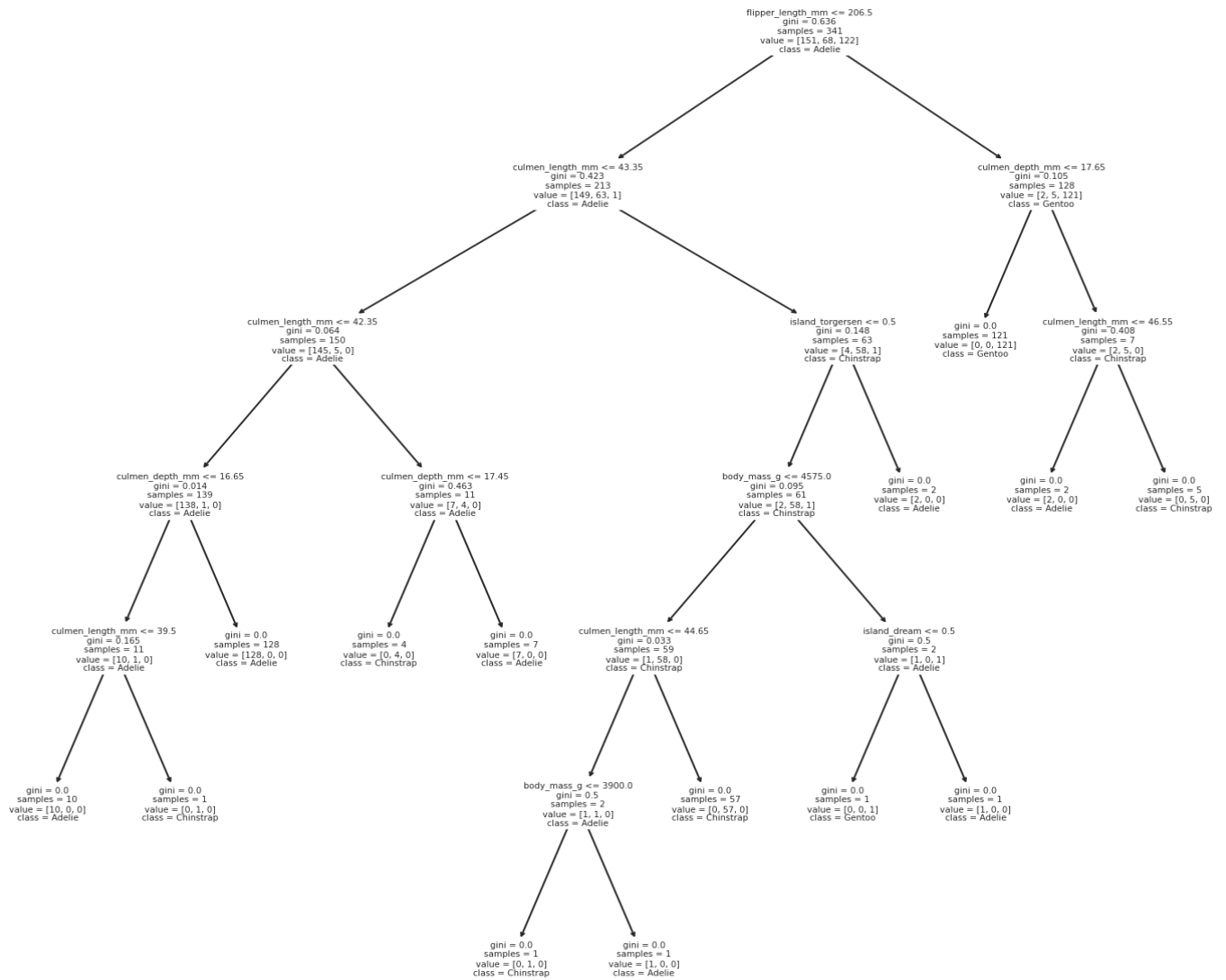
Provide the second two-part rule here.

**Here is my second simple two-part rule to identify gentoo:**

**If flipper length is greater than 206.5 mm and culmen depth is less than or equal to 17.65 mm, then the species is gentoo. Otherwise it is not.**



**Here is my second decision tree with labeled features and classes:**



### c. Positional encoding

Show the image obtained by predicting directly from  $(x,y)$  and the image obtained by predicting from the positional encoding.

**I am only claiming 10 points for this question because I was not able to create a multilayer network that predicts RGB from (x,y), or a multilayer network that predicts RGB from sinusoidal positional encoding of (x,y). However, I was able to recreate the image provided through an array of RGB values.**

Here is the step by step process:

1. I opened the provided image.
2. I converted the provided image to an array of RGB values.
3. I converted the array of RGB values to a new image (which matches the provided image).

Here is the recreated image (left), and the provided image (right). If you take a look at my code for this question, you can see the full process:



#### d. HW1 Survey

Don't need to put anything here. Just claim the points if (and only if) you submitted the HW1 survey by end of Feb 12.

#### Acknowledgments / Attribution

List any outside sources for code or improvement ideas or "None".

#### Sources:

1. Course Lectures/Slides
2. w3schools
3. [https://www.vertica.com/python/documentation\\_last/learn/DecisionTreeRegressor/](https://www.vertica.com/python/documentation_last/learn/DecisionTreeRegressor/)
4. [https://www.vertica.com/python/documentation\\_last/learn/RandomForestRegressor/](https://www.vertica.com/python/documentation_last/learn/RandomForestRegressor/)
5. <https://machinelearningmastery.com/gentle-introduction-gradient-boosting-algorithm-machine-learning/>
6. [https://pytorch.org/tutorials/beginner/deep\\_learning\\_60min\\_blitz.html](https://pytorch.org/tutorials/beginner/deep_learning_60min_blitz.html)
7. <https://www.simplilearn.com/tutorials/deep-learning-tutorial/multilayer-perceptron>
8. <https://towardsdatascience.com/master-positional-encoding-part-i-63c05d90a0c3>

9. <https://developers.google.com/machine-learning/decision-forests/decision-trees>
10. <https://machinelearningmastery.com/logistic-regression-for-machine-learning/>
11. <https://arxiv.org/pdf/2006.10739.pdf>

**Improvement Ideas:**

1. The only improvement I can think of is including more information that describes PyTorch.